

Text Classification using TF-IDF and Logistic Regression

Abstract: This report outlines the development and evaluation of a Natural Language Processing (NLP) pipeline designed for text classification. Built using scikit-learn, the model leverages a Term Frequency-Inverse Document Frequency (TF-IDF) vectorizer coupled with a Logistic Regression classifier. Through 5-fold cross-validation, the model achieved a mean F1 score of 0.6141.

1. Introduction

Text classification is a core NLP task focused on assigning predefined categories to text sequences. This report documents a machine learning approach to automate text categorization utilizing linear modeling applied to sparse vectorized text data.

2. Methodology

2.1 Data Preprocessing

The pipeline ingests training and testing sets provided in CSV format. The preprocessing involves:

- **Handling Missing Values:** Any empty text fields are replaced with empty strings to prevent pipeline failures.
- **Text Vectorization:** A TF-IDF Vectorizer transforms raw text into numerical feature arrays. Stop words are removed using the standard 'english' dictionary, and the vocabulary size is capped at a maximum of 10,000 features to control dimensionality and mitigate overfitting.

2.2 Model Architecture

The modeling strategy utilizes a standard scikit-learn Pipeline encapsulating two stages:

Stage	Component	Configuration
1. Vectorization	TfidfVectorizer	stop_words='english', max_features=10000
2. Classification	LogisticRegression	C=1.0, max_iter=1000, random_state=42

3. Training and Evaluation

The logistic regression model was trained on the processed `X_train` and `y_train` splits. To ensure robust performance estimates, the model was evaluated using 5-fold cross-validation.

Cross-Validation F1 Scores:

- Fold 1: 0.5882
- Fold 2: 0.5612
- Fold 3: 0.6048
- Fold 4: 0.5843
- Fold 5: 0.7321

Mean F1 Score: 0.6141

4. Conclusion and Inference

Following validation, the pipeline generated predictions on the unseen text sequences within `test.csv`. A two-column output mapping the test dataset's `id` to the predicted target classes was compiled. These final inferences were successfully saved to `submission.csv` for evaluation.